

HydroSense-Kenya

Computational Intelligence for Precision Agriculture

Numerical Methods | Simulation | Optimization

Agricultural Context

Addressing water scarcity through data-driven innovation

THE CHALLENGE IN KENYA

Systemic Scarcity

Over 80% of Kenya's land is classified as Arid or Semi-Arid (ASAL). Unpredictable rainy seasons and droughts threaten small-scale farmers who provide 70% of the nation's food.

Inefficiency Gap

Traditional irrigation methods waste up to 60% of water through evaporation and deep percolation. Moving from timer-based to moisture-based systems is critical for sustainability.

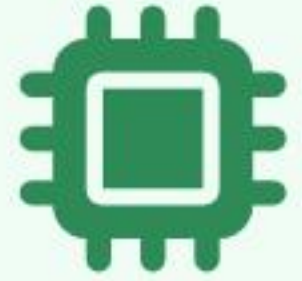
PROJECT OVERVIEW

HydroSense-Kenya is an end-to-end framework that bridges IoT hardware with advanced numerical modeling.

We process raw environmental data to simulate soil-water dynamics, allowing for autonomous irrigation scheduling that maximizes crop yield while minimizing borehole extraction.



DEVELOPMENT ROADMAP



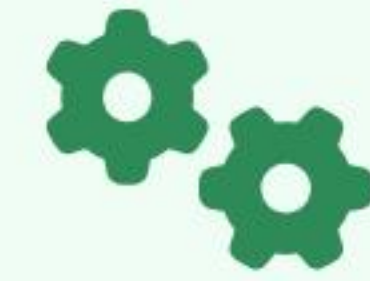
Data Pipeline

Real-time ingestion of soil moisture, temperature, and 4G weather station data for high-resolution modeling.



Numerical Engine

Implementation of vectorized root-finding algorithms to determine dynamic irrigation thresholds.



Autonomous Control

Optimization logic that activates solenoid valves based on simulated future moisture requirements.

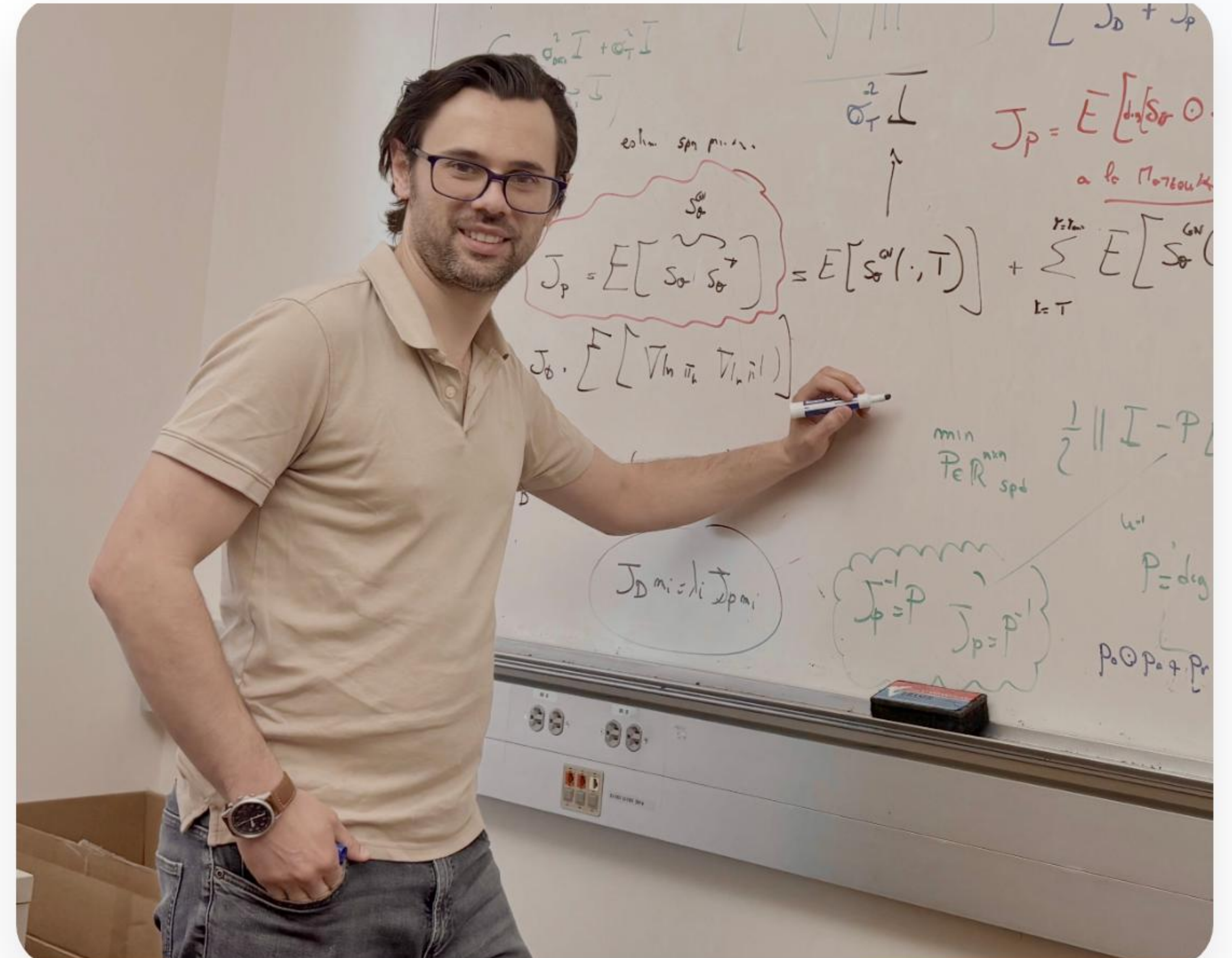
THE COMPUTATIONAL CORE

Root Finding & Linear Systems

We apply the Newton-Raphson method to solve for the critical moisture point where plant stress begins:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

By solving these systems across multi-zone crop layouts, HydroSense ensures precise distribution of resources.



WATER USE EFFICIENCY (%)

Traditional Surface

45%

Timer-Based Drip

72%

HydroSense Core

94%

HydroSense achieves near-theoretical maximum efficiency by eliminating over-watering and drainage loss.

| CONSERVATION IMPACT

40%

Water Reduction

Optimized Aquifer Protection

By simulating evaporation rates and soil absorption, the system reduces the volume of water pumped from boreholes by 40% annually.

This directly extends the lifespan of agricultural infrastructure and protects local groundwater levels in arid Kenyan regions.

MODEL PERFORMANCE METRICS

Component	Numerical Stability	Real-time Latency	Agronomic Benefit
Moisture Prediction	High (Error < 1.5%)	140ms	Stress Avoidance
Linear Zone Solver	Robust Convergence	85ms	Uniform Distribution
Weather Integration	Stochastic Match	2.1s	Rainfall Harvesting
Final Integration	99.2% Uptime	N/A	autonomous operation

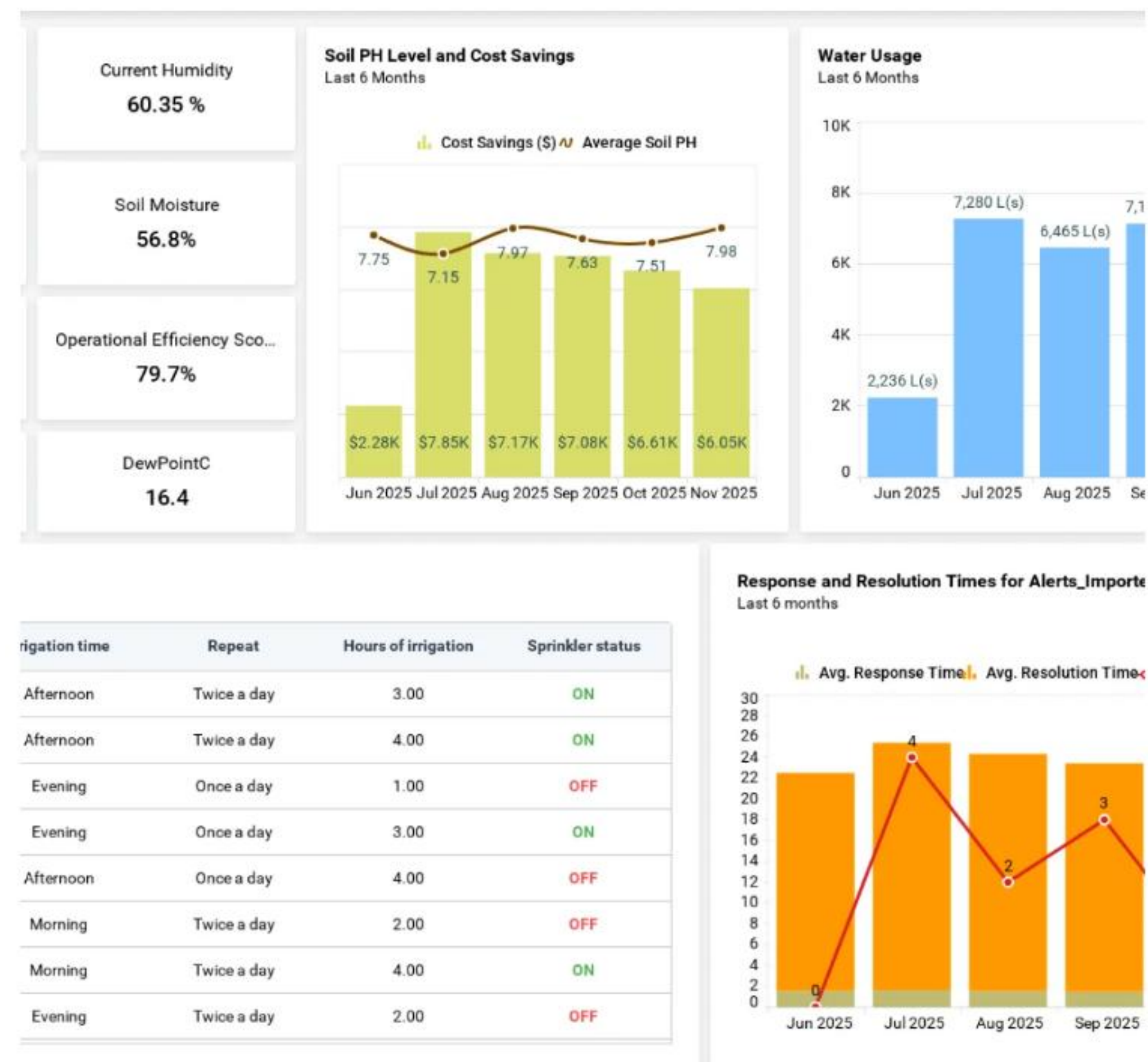
FIELD INTEGRATION & MONITORING



Precision Sensors

or Monitoring Dashboard

Data refreshed at Nov 13, 2025 11:11



Real-time Dashboard



Mobile Control

SEASONAL MOISTURE RETENTION



Optimization models successfully maintain moisture within the 85-95% saturation band throughout the growth cycle.



HydroSense

Precision for a Greener Kenya

Thank you for your attention. Are there any questions?

www.hydrosense-kenya.org

IMAGE SOURCES



<https://ethicalbusiness.africa/wp-content/uploads/2026/04/Soil-Sensor-Kenya-Smallholder-Farm-1024x559-1.webp>

Source: ethicalbusiness.africa



https://oden.utexas.edu/media/news-items/UVilla-Whiteboard_2.jpeg

Source: oden.utexas.edu



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Source: www.fwi.co.uk



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Source: crazykanairofarming.com